

Certificate Course in Basic Python[Self-Paced]

Chapter 1(Video 1)

Title:- Introduction to Python and Setting up Python environment for Jupyter Notebook

Question 1

What is an algorithm according to the video? a) A computer program written in Python b) A step-by-step procedure or formula for solving a problem c) A type of programming language d) A visual representation using symbols

Answer: b) A step-by-step procedure or formula for solving a problem

Question 2

What analogy does the instructor use to explain algorithms? a) Building a house b) Cooking recipe c) Reading a book d) Driving a car

Answer: b) Cooking recipe

Question 3

What is the primary reason computers need step-by-step instructions? a) They are faster than humans b) They can think independently c) They are dumb machines that cannot think by themselves d) They only work with Python

Answer: c) They are dumb machines that cannot think by themselves

Question 4

What sorting algorithm is demonstrated in the video? a) Quick Sort b) Merge Sort c) Bubble Sort d) Selection Sort

Answer: c) Bubble Sort

Question 5

In the bubble sort algorithm, what happens when two adjacent elements are in wrong order? a) The algorithm stops b) They are swapped c) They are deleted d) Nothing happens

Answer: b) They are swapped

Question 6

What type of programming language is Python according to the instructor? a) Compiled language b) Assembly language c) Interpreted language d) Machine language

Answer: c) Interpreted language

Question 7

What is a flowchart? a) A type of algorithm b) A programming language c) A visual representation of the sequence of steps and decisions needed to perform a process d) A Python function

Answer: c) A visual representation of the sequence of steps and decisions needed to perform a process

Question 8

What symbols are mentioned as being used in flowcharts? a) Circles and squares only b) Arrows, rectangles, and diamonds c) Only rectangles d) Numbers and letters

Answer: b) Arrows, rectangles, and diamonds

Question 9

What is the first step to use an algorithm effectively? a) Write the code immediately b) Choose a programming language c) Understand the problem you are trying to solve d) Draw a flowchart

Answer: c) Understand the problem you are trying to solve

Question 10

In which fields are algorithms used according to the video? a) Only in computer science b) Computer science, mathematics, data analysis, artificial intelligence, and daily activities c) Only in Python programming d) Only in software development

Answer: b) Computer science, mathematics, data analysis, artificial intelligence, and daily activities

Answer Key

1. b
2. b
3. c
4. c
5. b
6. c
7. c
8. b
9. c
10. b

chapter 2 (video 2)

Title:- Basic syntax and data types

Question 1: Python Installation Requirement

What important checkbox must be checked during Python installation according to the instructor?

- a) Install Python documentation
- b) Add Python to environment variables
- c) Create desktop shortcut
- d) Install pip package manager

Answer: b) Add Python to environment variables

Question 2: Code Editor Choice

Which code editor does the instructor recommend and personally use for Python development?

- a) PyCharm
- b) Sublime Text
- c) Visual Studio Code
- d) Atom

Answer: c) Visual Studio Code

Question 3: Essential VS Code Extensions

Which extensions does the instructor mention as essential for Python development in VS Code?

- a) Python and GitLens
- b) Python and Code Runner
- c) Code Runner and Prettier
- d) Python and Live Server

Answer: b) Python and Code Runner

Question 4: Python File Extension

What file extension should always be used for Python files according to the instructor?

- a) .txt
- b) .py
- c) .python
- d) .pyt

Answer: b) .py

Question 5: Python Interpreter Configuration

How do you configure the Python interpreter in VS Code according to the demonstration?

- a) Go to File menu and select Python Interpreter
- b) Press Ctrl+Shift+P and search for "Python: Select Interpreter"
- c) Right-click on any Python file and select interpreter
- d) Go to Settings and search for Python

Answer: b) Press Ctrl+Shift+P and search for "Python: Select Interpreter"

Answer Key:

- 1. **b** - Environment variables checkbox is crucial
 - 2. **c** - Visual Studio Code is the recommended editor
 - 3. **b** - Python and Code Runner extensions are essential
 - 4. **b** - .py extension is required for Python files
 - 5. **b** - Ctrl+Shift+P opens command palette for interpreter selection
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Question 6: Installation Source

From where should Python be downloaded according to the instructor?

- a) Third-party software websites
- b) Microsoft Store
- c) The official Python website
- d) GitHub repositories

Answer: c) The official Python website

Question 7: Terminal Access

What keyboard shortcut opens the integrated terminal in VS Code?

- a) Ctrl + T
- b) Ctrl + `
- c) Ctrl + Shift + T
- d) Alt + T

Answer: b) Ctrl + `

Question 8: Project Organization

What does the instructor recommend for organizing Python projects in VS Code?

- a) Keep all files in the desktop
- b) Create separate folders for each project and open them in VS Code
- c) Store everything in the Documents folder
- d) Use the default VS Code workspace

Answer: b) Create separate folders for each project and open them in VS Code

Question 9: Bonus Extension

Which additional extension does the instructor mention as a "bonus tip"?

- a) Prettier
- b) GitLens
- c) Auto Save
- d) Live Server

Answer: c) Auto Save

Question 10: Course Progression

What does the instructor announce will happen from the next session onwards?

- a) More theoretical concepts about programming
- b) Advanced algorithm discussions
- c) Actual Python coding will begin
- d) Installation of more software tools

Answer: c) Actual Python coding will begin

Complete Answer Key:

1. **b** - Environment variables checkbox is crucial
 2. **c** - Visual Studio Code is the recommended editor
 3. **b** - Python and Code Runner extensions are essential
 4. **b** - .py extension is required for Python files
 5. **b** - Ctrl+Shift+P opens command palette for interpreter selection
 6. **c** - Official Python website for downloads
 7. **b** - Ctrl + ` opens integrated terminal
 8. **b** - Organize projects in separate folders
 9. **c** - Auto Save extension as bonus tip
 10. **c** - Actual coding starts from next session
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Additional Setup Notes:

- **Download Python from:** Official Python website (python.org)
- **Bonus Extension:** Auto Save (mentioned by instructor)
- **Terminal Access:** Ctrl + ` opens integrated terminal
- **File Creation:** Always save Python files with .py extension
- **Virtual Environment:** Recommended for project isolation (mentioned but not detailed)
- **Pylint Installation:** Recommended for code linting and error detection

Chapter :- 3

Title :- Conditional statements

Question 1

What is the instructor's preferred code editor for Python development? a) PyCharm b) Visual Studio Code c) Sublime Text d) Atom

Answer: b) Visual Studio Code

Question 2

Which Python version was mentioned for download in the video? a) Python 3.11.2 b) Python 3.12.5 c) Python 3.10.4 d) Python 2.7.18

Answer: b) Python 3.12.5

Question 3

What is the most important checkbox to check during Python installation? a) Install pip b) Install documentation c) Add Python to environment variables d) Create desktop shortcut

Answer: c) Add Python to environment variables

Question 4

From which website should you download Python? a) Any third-party website b) The official Python website c) GitHub d) Microsoft Store

Answer: b) The official Python website

Question 5

What are the three essential extensions mentioned for VS Code when working with Python? a) Python, Code Runner, and Auto Save b) Python, Debugger, and Linter c) Python, Git, and Auto Save d) Code Runner, Debugger, and Auto Save

Answer: a) Python, Code Runner, and Auto Save

Question 6

What keyboard shortcut is used to configure the Python interpreter in VS Code? a) Ctrl + Shift + I b) Ctrl + Shift + P c) Ctrl + Alt + P d) Ctrl + P

Answer: b) Ctrl + Shift + P

Question 7

What file extension must be used for Python files? a) .txt b) .py c) .python d) .pyt

Answer: b) .py

Question 8

What keyboard shortcut opens the terminal in VS Code? a) Ctrl + T b) Ctrl + ` c) Ctrl + Shift + T d) Alt + T

Answer: b) Ctrl + `

Question 9

What should you do after installing VS Code to make Python development easier? a) Install Python extension only b) Configure the Python interpreter and install necessary extensions c) Just start coding immediately d) Install Git first

Answer: b) Configure the Python interpreter and install necessary extensions

Question 10

What does the instructor recommend for project isolation? a) Using different computers b) Creating separate folders c) Setting up a virtual environment d) Using different Python versions

Answer: c) Setting up a virtual environment

Answer Key

1. b
2. b
3. c
4. b
5. a
6. b
7. b
8. b
9. b
10. c

Chapter 4

Title :- Loops in Python

Question 1

What is syntax in programming according to the video? a) A type of programming language b) The set of rules that dictate how code must be written to be understood by the compiler or interpreter c) A method of debugging code d) A way to optimize code performance

Answer: b) The set of rules that dictate how code must be written to be understood by the compiler or interpreter

Question 2

Which symbol is used for single-line comments in Python? a) // b) /* c) # d) --

Answer: c) #

Question 3

What are the four basic data types mentioned in the video? a) Integers, decimals, text, and boolean b) Numbers, strings, lists, and dictionaries c) Integers, floats, strings, and booleans d) Whole numbers, fractions, words, and logic

Answer: c) Integers, floats, strings, and booleans

Question 4

Which data type would you use to store a person's age? a) Float b) String c) Boolean d) Integer

Answer: d) Integer

Question 5

What are the only two possible values for a boolean data type? a) Yes and No b) True and False c) 1 and 0 d) On and Off

Answer: b) True and False

Question 6

Which rule is correct for Python variable names? a) Variable names can start with numbers b) Variable names must start with a letter or underscore character c) Variable names can contain spaces d) Variable names are not case sensitive

Answer: b) Variable names must start with a letter or underscore character

Question 7

What function is used to determine the data type of a variable in Python? a) datatype() b) typeof() c) type() d) gettype()

Answer: c) type()

Question 8

What is the process of converting one data type to another called? a) Transformation b) Casting c) Conversion d) Morphing

Answer: b) Casting

Question 9

Which of the following is NOT a valid Python variable naming convention mentioned in the video? a) camelCase b) PascalCase c) snake_case d) kebab-case

Answer: d) kebab-case

Question 10

Why can't you use Python keywords as variable names? a) It makes the code run slower b) It causes memory issues c) Keywords are reserved words with special meanings in Python d) It makes the code harder to read

Answer: c) Keywords are reserved words with special meanings in Python

Answer Key

1. b
2. c
3. c
4. d
5. b
6. b
7. c
8. b
9. d
10. c

Chapter 5

Title:- Functions for Python

Question 1

What is the primary purpose of conditional statements in Python? a) To store data in variables b) To make decisions based on certain conditions c) To perform mathematical calculations d) To create functions

Answer: b) To make decisions based on certain conditions

Question 2

What are the three main conditional statements in Python mentioned in the video? a) if, then, else b) if, elif, else c) when, then, otherwise d) check, verify, default

Answer: b) if, elif, else

Question 3

In the age example, what operator was used to check if age is greater than or equal to 18? a) > b) == c) >= d) <=

Answer: c) >=

Question 4

What happens when an if condition is false and there's no else statement? a) The program crashes b) An error is thrown c) Nothing gets executed and the program continues d) A default message is printed

Answer: c) Nothing gets executed and the program continues

Question 5

In the grading system example with score = 85, what would be printed? a) Grade A b) Grade B c) Grade C d) No grade

Answer: b) Grade B

Question 6

In the e-commerce discount example, if amount_spent = 150, what discount would be applied? a) 5% b) 10% c) 20% d) 0%

Answer: b) 10%

Question 7

What is the difference between 'if' and 'elif' statements? a) There is no difference b) elif tests another condition if the previous if condition was false c) elif is used for mathematical operations only d) elif can only be used once in a program

Answer: b) elif tests another condition if the previous if condition was false

Question 8

When does the 'else' statement execute? a) Always after the if statement b) Only when the if condition is true c) When none of the preceding conditions were true d) Before the if statement

Answer: c) When none of the preceding conditions were true

Question 9

What are the two main types of loops mentioned in Python? a) repeat and continue b) for and while c) loop and cycle d) iterate and repeat

Answer: b) for and while

Question 10

What is a for loop primarily used for according to the video? a) To create variables b) To iterate over a sequence like a list, tuple, dictionary, set, or string c) To perform mathematical calculations d) To handle errors in code

Answer: b) To iterate over a sequence like a list, tuple, dictionary, set, or string

Answer Key

1. b
2. b
3. c
4. c
5. b
6. b
7. b
8. c
9. b
10. b

chapter 6

Title :- Data Structures

video 1

Question 1

When is a for loop typically used according to the video? a) When you need to make decisions b) When you need to iterate over a collection of items or a range of numbers c) When you need to store data d) When you need to create functions

Answer: b) When you need to iterate over a collection of items or a range of numbers

Question 2

When is a while loop used? a) When the number of iterations is predetermined b) When you need to iterate over a list c) When the number of iterations is not predetermined and depends on a condition d) When you need to print something once

Answer: c) When the number of iterations is not predetermined and depends on a condition

Question 3

What is the correct syntax for a for loop in Python? a) for variable in iterable: b) for (variable in iterable): c) loop variable in iterable: d) iterate variable in iterable:

Answer: a) for variable in iterable:

Question 4

In the example `for i in range(1, 10)`, which numbers will be printed? a) 1 to 10 (including 10) b) 1 to 9 (excluding 10) c) 0 to 9 d) 1 to 8

Answer: b) 1 to 9 (excluding 10)

Question 5

What does the `range()` function do in a for loop? a) It creates a list of strings b) It generates numbers dynamically within a specified range c) It counts the number of items in a list d) It stops the loop execution

Answer: b) It generates numbers dynamically within a specified range

Question 6

In the while loop example with `count = 1`, what does `count += 1` mean? a) `count = 1` b) `count = count + 1` c) `count = count - 1` d) `count = 0`

Answer: b) `count = count + 1`

Question 7

What function is used to find the length of a string or variable? a) length() b) size() c) len() d) count()

Answer: c) len()

Question 8

In the e-commerce products example, what was the purpose of using a for loop? a) To create new products b) To delete products from inventory c) To display all product names in the inventory d) To count the number of products

Answer: c) To display all product names in the inventory

Question 9

In the password validation example, what condition keeps the while loop running? a) While password is accepted b) While length of password is greater than 8 c) While length of password is less than 8 d) While password is entered

Answer: c) While length of password is less than 8

Question 10

According to the summary, what is the main difference between for loops and while loops? a) For loops are faster than while loops b) For loops are ideal when iterations are known beforehand, while loops are flexible for dynamic conditions c) While loops can only work with numbers d) For loops can only work with strings

Answer: b) For loops are ideal when iterations are known beforehand, while loops are flexible for dynamic conditions

Answer Key

1. b
2. c
3. a
4. b
5. b
6. b
7. c
8. c
9. c
10. b

video 2

Question 1

What is a function in Python according to the video? a) A type of variable b) A block of reusable code designed to perform a specific task c) A way to store data d) A method to create loops

Answer: b) A block of reusable code designed to perform a specific task

Question 2

What keyword is used to define a function in Python? a) function b) define c) def d) func

Answer: c) def

Question 3

Why are functions used in programming? a) To make programs run faster b) To avoid code duplication and manage complexity c) To create variables d) To handle user input only

Answer: b) To avoid code duplication and manage complexity

Question 4

What is the correct syntax for defining a function in Python? a) `def function_name():` b) `function function_name():` c) `define function_name():` d) `func function_name():`

Answer: a) `def function_name():`

Question 5

How do you call a function that has been defined? a) By writing the function definition again b) By using the function name followed by parentheses c) By using the call keyword d) By importing the function

Answer: b) By using the function name followed by parentheses

Question 6

What are parameters in a function? a) The output of the function b) Variables specified within the parentheses of the function definition c) The function name d) Comments in the function

Answer: b) Variables specified within the parentheses of the function definition

Question 7

What does the `return` statement do in a function? a) It prints the result to the console b) It ends the program c) It allows the function to return a value for further use d) It creates a loop

Answer: c) It allows the function to return a value for further use

Question 8

In the example `greet("Harish")`, what is "Harish"? a) A parameter b) An argument c) A function name d) A return value

Answer: b) An argument

Question 9

What is the difference between `print()` and `return` in a function? a) There is no difference b) Print displays output to console, return allows capturing output for further use c) Return is faster than print d) Print can only be used once

Answer: b) Print displays output to console, return allows capturing output for further use

Question 10

In the circle area calculation example, which library was imported? a) numpy b) pandas c) math d) random

Answer: c) math

Answer Key

1. b
2. c
3. b
4. a
5. b
6. b
7. c
8. b
9. b
10. c

video 3

Question 1

What is a function in Python according to the video? a) A type of variable b) A block of reusable code designed to perform a specific task c) A way to store data d) A method to create loops

Answer: b) A block of reusable code designed to perform a specific task

Question 2

What keyword is used to define a function in Python? a) function b) define c) def d) func

Answer: c) def

Question 3

Why are functions used in programming? a) To make programs run faster b) To avoid code duplication and manage complexity c) To create variables d) To handle user input only

Answer: b) To avoid code duplication and manage complexity

Question 4

What is the correct syntax for defining a function in Python? a) def function_name(): b) function function_name(): c) define function_name(): d) func function_name():

Answer: a) def function_name():

Question 5

How do you call a function that has been defined? a) By writing the function definition again b) By using the function name followed by parentheses c) By using the call keyword d) By importing the function

Answer: b) By using the function name followed by parentheses

Question 6

What are parameters in a function? a) The output of the function b) Variables specified within the parentheses of the function definition c) The function name d) Comments in the function

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Question 8

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Question 10

In the circle area calculation example, which library was imported? a) numpy b) pandas c) math d) random

Answer: c) math

Answer Key

1. b
2. c
3. b
4. a
5. b
6. b
7. c
8. b
9. b
10. c

video (4)

Question 1

What are the main advantages of Python lists mentioned in the video? a) They are fast and consume less memory b) Flexibility, ease of use, order preservation, efficient access, and versatility c) They can only store numbers d) They are immutable and secure

Answer: b) Flexibility, ease of use, order preservation, efficient access, and versatility

Question 2

Which method is used to add an element to the end of a list? a) add() b) insert() c) append() d) push()

Answer: c) append()

Question 3

Which method is used to remove an element from a list? a) delete() b) remove() c) pop() d) Both b and c

Answer: d) Both b and c

Question 4

What principle does a stack follow when implemented using a list? a) First In First Out (FIFO)
b) Last In First Out (LIFO) c) Random Access d) Priority Based

Answer: b) Last In First Out (LIFO)

Question 5

In the stack implementation shown, what does the `pop()` method do by default? a) Removes the first element b) Removes a random element c) Removes and returns the last element d) Removes all elements

Answer: c) Removes and returns the last element

Question 6

How do you access the first element of a list named `cart`? a) `cart[1]` b) `cart[0]` c) `cart.first()` d) `cart[-1]`

Answer: b) `cart[0]`

Question 7

What does `range(10)` generate in Python? a) Numbers from 1 to 10 b) Numbers from 0 to 10 c) Numbers from 0 to 9 d) Numbers from 1 to 9

Answer: c) Numbers from 0 to 9

Question 8

In the expression `i ** 2`, what does the `**` operator represent? a) Multiplication b) Division c) Exponentiation (power) d) Modulus

Answer: c) Exponentiation (power)

Question 9

What is list comprehension? a) A way to delete lists b) A concise way to create lists based on existing lists or ranges c) A method to sort lists d) A way to convert lists to strings

Answer: b) A concise way to create lists based on existing lists or ranges

Question 10

Which of the following best describes how lists can be used for dynamic data handling? a) Lists have fixed size and cannot be modified b) Lists can be dynamically modified by adding and removing items during runtime c) Lists can only store one data type d) Lists cannot be used in loops

Answer: b) Lists can be dynamically modified by adding and removing items during runtime

Answer Key

1. b
2. c
3. d
4. b
5. c
6. b
7. c
8. c
9. b
10. b

Chapter 7

Title:- File Handling in Python

video 1

Question 1

What is the syntax for slicing a list in Python? a) list{start:stop:step} b) list[start:stop:step] c) list(start:stop:step) d) list[start:stop:step](#)

Answer: b) list[start:stop:step]

Question 2

In list slicing, what happens to the 'stop' index? a) It is included in the slice b) It is excluded from the slice c) It becomes the first element d) It causes an error

Answer: b) It is excluded from the slice

Question 3

What are the default values for start, stop, and step in list slicing? a) start=1, stop=length, step=1 b) start=0, stop=length, step=1 c) start=0, stop=length-1, step=1 d) start=1, stop=length-1, step=2

Answer: b) start=0, stop=length, step=1

Question 4

How do you reverse a list using slicing? a) list[::-1] b) list[::1] c) list[:-1:] d) list[1::-1]

Answer: a) list[::-1]

Question 5

What does the `append()` method do? a) Adds an element at the beginning of the list b) Adds an element at the end of the list c) Adds an element at a specific index d) Removes an element from the list

Answer: b) Adds an element at the end of the list

Question 6

What is the difference between `append()` and `insert()` methods? a) There is no difference b) `append()` adds at the end, `insert()` adds at a specific index c) `append()` is faster than `insert()` d) `insert()` only works with numbers

Answer: b) `append()` adds at the end, `insert()` adds at a specific index

Question 7

What does the `remove()` method do when there are duplicate elements? a) Removes all occurrences of the element b) Removes the last occurrence of the element c) Removes the first occurrence of the element d) Throws an error

Answer: c) Removes the first occurrence of the element

Question 8

What happens when you call `pop()` without any arguments? a) It removes the first element b) It removes the last element c) It removes a random element d) It throws an error

Answer: b) It removes the last element

Question 9

What does the `sort()` method do to a list? a) Sorts the list in descending order b) Sorts the list in ascending order c) Sorts the list randomly d) Creates a new sorted list

Answer: b) Sorts the list in ascending order

Question 10

If you have a list `fruits = ['banana', 'papaya', 'cherry', 'mango']`, what will `fruits[1:3]` return? a) `['banana', 'papaya']` b) `['papaya', 'cherry']` c) `['papaya', 'cherry', 'mango']` d) `['banana', 'papaya', 'cherry']`

Answer: b) `['papaya', 'cherry']`

Answer Key

1. b
2. b
3. b
4. a
5. b
6. b
7. c
8. b
9. b
10. b

Chapter 8

Title:- Modules and Packages in Python

video 1

Question 1

What is the key characteristic of tuples in Python? a) They are mutable and can be changed after creation b) They are immutable and cannot be changed after creation c) They can only store integers d) They are faster than all other data structures

Answer: b) They are immutable and cannot be changed after creation

Question 2

How do you create a tuple in Python? a) Using square brackets [] b) Using curly braces {} c) Using parentheses () with elements separated by commas d) Using angle brackets <>

Answer: c) Using parentheses () with elements separated by commas

Question 3

What is required when creating a tuple with a single element? a) Nothing special is required b) You must include a comma after the element c) You must use double parentheses d) You cannot create a single element tuple

Answer: b) You must include a comma after the element

Question 4

Can you create a tuple without parentheses? a) No, parentheses are always required b) Yes, you can use just commas to separate elements c) Only if the tuple has more than 3 elements d) Only with numeric data

Answer: b) Yes, you can use just commas to separate elements

Question 5

How do you access elements in a tuple? a) Using indexing similar to lists, starting from 0 b) Using the `get()` method c) Using square brackets with element names d) Tuples cannot be accessed directly

Answer: a) Using indexing similar to lists, starting from 0

Question 6

What does negative indexing in tuples allow you to do? a) Create negative numbers in tuples b) Access elements from the end of the tuple c) Remove elements from the tuple d) Sort the tuple in reverse order

Answer: b) Access elements from the end of the tuple

Question 7

What does the `count()` method do in tuples? a) Counts the total number of elements in the tuple b) Returns the number of times a specified value appears in the tuple c) Counts the number of different data types in the tuple d) Counts the memory usage of the tuple

Answer: b) Returns the number of times a specified value appears in the tuple

Question 8

What does the `index()` method return in tuples? a) All positions where an element appears b) The last occurrence of an element c) The index of the first occurrence of a specified value d) The total count of elements

Answer: c) The index of the first occurrence of a specified value

Question 9

What is tuple packing? a) Compressing tuples to save memory b) Assigning multiple values to a tuple at once c) Converting tuples to lists d) Removing elements from tuples

Answer: b) Assigning multiple values to a tuple at once

Question 10

What is tuple unpacking? a) Deleting all elements from a tuple b) Converting tuples to other data types c) Extracting values from a tuple into individual variables d) Sorting elements in a tuple

Answer: c) Extracting values from a tuple into individual variables

Answer Key

1. b
2. c
3. b
4. b
5. a
6. b
7. b
8. c
9. b
10. c

video 2

Question 1

What is a dictionary in Python? a) An ordered, immutable collection of elements b) An unordered, mutable collection that stores data in key-value pairs c) A collection that only stores strings d) An ordered collection similar to lists

Answer: b) An unordered, mutable collection that stores data in key-value pairs

Question 2

How do you create a dictionary in Python? a) Using square brackets [] b) Using parentheses () c) Using curly braces {} with key-value pairs d) Using angle brackets <>

Answer: c) Using curly braces {} with key-value pairs

Question 3

What are the requirements for dictionary keys? a) Keys must be mutable and can be duplicated b) Keys must be unique and immutable (like strings, numbers, or tuples) c) Keys can only be integers d) Keys must be strings only

Answer: b) Keys must be unique and immutable (like strings, numbers, or tuples)

Question 4

What types of data can dictionary values be? a) Only strings and numbers b) Only immutable data types c) Any data type including numbers, strings, lists, or even other dictionaries d) Only the same type as the key

Answer: c) Any data type including numbers, strings, lists, or even other dictionaries

Question 5

How do you access a value in a dictionary? a) Using the index number like lists b) Using square brackets with the key name: dict[key] c) Using the get() method only d) Using parentheses with the key name

Answer: b) Using square brackets with the key name: dict[key]

Question 6

How do you add a new key-value pair to an existing dictionary? a) Using the `append()` method b) Using the `insert()` method c) By assigning a value to a new key: `dict[new_key] = value` d) Using the `add()` method

Answer: c) By assigning a value to a new key: `dict[new_key] = value`

Question 7

What does the `pop()` method do in dictionaries? a) Removes the last added item b) Removes and returns the value of the item with the specified key c) Removes the first item d) Removes all items

Answer: b) Removes and returns the value of the item with the specified key

Question 8

What does the `del` keyword do when used with dictionaries? a) Deletes the entire dictionary b) Deletes the last item c) Deletes the item with the specified key d) Deletes all values but keeps keys

Answer: c) Deletes the item with the specified key

Question 9

What does the `popitem()` method do? a) Removes and returns a random key-value pair b) Removes the first inserted key-value pair c) Removes the last inserted key-value pair d) Removes all items

Answer: c) Removes the last inserted key-value pair

Question 10

What does the `clear()` method do to a dictionary? a) Removes only the values, keeping the keys b) Removes only the keys, keeping the values c) Removes all items from the dictionary, making it empty d) Sorts the dictionary

Answer: c) Removes all items from the dictionary, making it empty

Answer Key

1. b
2. c
3. b
4. c
5. b
6. c
7. b
8. c
9. c
10. c

Chapter 9

Title:- Object Oriented Programming(OOPs)

video 1

Question 1

What does the `fromkeys()` method do in dictionaries? a) Creates a copy of an existing dictionary b) Creates a new dictionary from specified keys with each key assigned the same value c) Removes keys from a dictionary d) Sorts the dictionary by keys

Answer: b) Creates a new dictionary from specified keys with each key assigned the same value

Question 2

What does the `update()` method do in dictionaries? a) Changes the order of dictionary items b) Updates the dictionary with key-value pairs from another dictionary c) Updates only the values, keeping keys unchanged d) Creates a backup of the dictionary

Answer: b) Updates the dictionary with key-value pairs from another dictionary

Question 3

What is a set in Python? a) An ordered collection that allows duplicate elements b) An unordered collection of unique elements that does not allow duplicates c) A mutable collection that preserves insertion order d) An immutable collection similar to tuples

Answer: b) An unordered collection of unique elements that does not allow duplicates

Question 4

How do you create a set in Python? a) Using square brackets [] b) Using parentheses () c) Using curly braces {} or the set() function d) Using angle brackets <>

Answer: c) Using curly braces {} or the set() function

Question 5

How do you create an empty set in Python? a) Using empty curly braces {} b) Using the set() function c) Using empty square brackets [] d) Using empty parentheses ()

Answer: b) Using the set() function

Question 6

How can you access elements in a set? a) Using index numbers like lists b) Using keys like dictionaries c) By looping through the set or checking if an item exists using 'in' d) Sets cannot be accessed

Answer: c) By looping through the set or checking if an item exists using 'in'

Question 7

What method is used to add an element to a set? a) append() b) insert() c) add() d) push()

Answer: c) add()

Question 8

What method is used to remove an element from a set? a) delete() b) remove() or discard() c) pop() only d) clear() only

Answer: b) remove() or discard()

Question 9

What is the difference between `remove()` and `discard()` methods in sets? a) There is no difference b) `remove()` throws an error if element doesn't exist, `discard()` doesn't c) `discard()` is faster than `remove()` d) `remove()` can remove multiple elements, `discard()` cannot

Answer: b) remove() throws an error if element doesn't exist, discard() doesn't

Question 10

What are the key characteristics of sets mentioned in the video? a) Ordered, immutable, allows duplicates b) Unordered, unique elements, mutable (but elements must be immutable) c) Ordered, unique elements, immutable d) Unordered, allows duplicates, immutable

Answer: b) Unordered, unique elements, mutable (but elements must be immutable)

Answer Key

1. b
2. b
3. b
4. c
5. b
6. c
7. c

- 8. b
- 9. b
- 10. b

video 2

Question 1

What does the `pop()` method do when used with sets? a) Removes the first element b) Removes the last element c) Removes a random element and returns it d) Removes all elements

Answer: c) Removes a random element and returns it

Question 2

What is the difference between `remove()` and `discard()` methods in sets? a) There is no difference b) `remove()` raises an error if the element doesn't exist, `discard()` doesn't c) `discard()` is faster than `remove()` d) `remove()` can remove multiple elements

Answer: b) `remove()` raises an error if the element doesn't exist, `discard()` doesn't

Question 3

What does the `del` keyword do when used with sets? a) Removes a specific element b) Removes all elements but keeps the set c) Deletes the entire set completely d) Removes the last element

Answer: c) Deletes the entire set completely

Question 4

What does the `union()` method do with sets? a) Removes duplicate elements from a single set b) Returns a new set with all items from both sets c) Finds only the common elements d) Updates the original set

Answer: b) Returns a new set with all items from both sets

Question 5

Which operator can be used as an alternative to the `union()` method? a) `&` (ampersand) b) `|` (pipe/pipeline) c) `+` (plus) d) `-` (minus)

Answer: b) `|` (pipe/pipeline)

Question 6

What does the `intersection()` method return? a) All elements from both sets b) Elements that are only in the first set c) A new set containing only items present in both sets d) Elements that are only in the second set

Answer: c) A new set containing only items present in both sets

Question 7

Which operator can be used as an alternative to the `intersection()` method? a) `|` (pipe) b) `&` (ampersand) c) `+` (plus) d) `-` (minus)

Answer: b) `&` (ampersand)

Question 8

What is the key difference between `union()` and `update()` methods? a) There is no difference b) `union()` creates a new set, `update()` modifies the original set c) `update()` is faster than `union()` d) `union()` only works with two sets

Answer: b) `union()` creates a new set, `update()` modifies the original set

Question 9

Can you join sets with other data types like tuples using the union method? a) No, only sets can be joined b) Yes, but the result will always be a set c) Yes, and the result will be a tuple d) Only with lists, not tuples

Answer: b) Yes, but the result will always be a set

Question 10

When iterating through a set using a for loop, what can you expect about the order of elements?

a) Elements will always appear in the order they were added b) Elements will appear in alphabetical/numerical order c) The order is random and may vary each time d) Elements will appear in reverse order

Answer: c) The order is random and may vary each time

Answer Key

1. c
2. b
3. c
4. b
5. b
6. c
7. b
8. b
9. b
10. c

video 3

Question 1

What are comprehensions in Python? a) A way to compress files b) A concise way to create new sequences from existing sequences using a single line of code c) A method to understand complex code d) A debugging technique

Answer: b) A concise way to create new sequences from existing sequences using a single line of code

Question 2

What is the basic syntax for list comprehension? a) [expression if condition for item in iterable] b) [expression for item in iterable if condition] c) [for item in iterable: expression] d) [item expression iterable condition]

Answer: b) [expression for item in iterable if condition]

Question 3

In list comprehension syntax, what does the "expression" represent? a) The condition to filter items b) The iterable object c) The value or operation applied to each item d) The variable name

Answer: c) The value or operation applied to each item

Question 4

What does this list comprehension create: `[x * x for x in range(1, 6)]`? a) A list of numbers from 1 to 6 b) A list of squares of numbers from 1 to 5 c) A list of numbers multiplied by themselves twice d) A list of even numbers

Answer: b) A list of squares of numbers from 1 to 5

Question 5

In the list comprehension `[x for x in range(2, 12) if x % 2 == 0]`, what does `x % 2 == 0` check for? a) Odd numbers b) Even numbers c) Numbers divisible by 12 d) Prime numbers

Answer: b) Even numbers

Question 6

What is the purpose of the `if condition` part in list comprehension? a) To define the loop variable b) To specify the iterable c) To filter items based on a condition d) To define the expression

Answer: c) To filter items based on a condition

Question 7

What is the basic syntax for dictionary comprehension? a) {key: value for item in iterable} b) {key_expression: value_expression for item in iterable} c) [key: value for item in iterable] d) (key, value for item in iterable)

Answer: b) {key_expression: value_expression for item in iterable}

Question 8

What does this dictionary comprehension create: {x: x*x for x in range(1, 6)}? a) A dictionary with numbers as keys and their squares as values b) A dictionary with squares as keys and numbers as values c) A list of key-value pairs d) A set of numbers and their squares

Answer: a) A dictionary with numbers as keys and their squares as values

Question 9

What are the main advantages of using comprehensions over traditional loops? a) They are easier to debug b) They are faster and more memory efficient c) They can handle more complex conditions d) They work with more data types

Answer: b) They are faster and more memory efficient

Question 10

Which of the following is NOT a type of comprehension mentioned in the video? a) List comprehension b) Dictionary comprehension c) Set comprehension d) Tuple comprehension

Answer: d) Tuple comprehension

Answer Key

1. b
2. b
3. c
4. b
5. b
6. c
7. b
8. a
9. b
10. d

Chapter 10

Title :- Exceptions & Error Handling

video 1

Question 1

What is file management in Python? a) A way to organize Python code b) Working with files and directories to read, write, append, delete and modify them c) Managing memory usage in Python programs d) Organizing variables in Python

Answer: b) Working with files and directories to read, write, append, delete and modify them

Question 2

Which function is primarily used to open files in Python? a) read() b) file() c) open() d) access()

Answer: c) open()

Question 3

What are the two main arguments that the `open()` function takes? a) File size and file type b) File name and file mode c) File path and file content d) File extension and file permissions

Answer: b) File name and file mode

Question 4

What does the file mode 'r' represent? a) Remove b) Read (default mode) c) Replace d) Reverse

Answer: b) Read (default mode)

Question 5

What does the file mode 'w' do when opening a file? a) Writes to the end of the file b) Writes only if the file doesn't exist c) Overwrites the file if it exists d) Creates a backup before writing

Answer: c) Overwrites the file if it exists

Question 6

What does the file mode 'a' represent? a) Access - allows reading the file b) Append - adds data to the end of the file c) Archive - compresses the file d) Auto - automatically detects the mode

Answer: b) Append - adds data to the end of the file

Question 7

Why is it important to close a file after opening it? a) To save memory and release system resources b) To make the file readable by other programs c) To prevent the file from being corrupted d) All of the above

Answer: d) All of the above

Question 8

What is the advantage of using the `with` keyword when working with files? a) It makes the code run faster b) It automatically closes the file when done c) It prevents file corruption d) It allows multiple files to be opened simultaneously

Answer: b) It automatically closes the file when done

Question 9

What method is used to read the content of a file? a) `file.get()` b) `file.content()` c) `file.read()` d) `file.load()`

Answer: c) file.read()

Question 10

Which approach is better for reading large files? a) Reading the entire file at once using read() b) Reading the file line by line using a loop c) Loading the file into memory first d) Using multiple threads to read different parts

Answer: b) Reading the file line by line using a loop

Answer Key

1. b
2. c
3. b
4. b
5. c
6. b
7. d
8. b
9. c
10. b

video 2

Question 1

What is a module in Python? a) A built-in function b) A file containing Python code with functions, classes, and variables c) A type of variable d) A Python keyword

Answer: b) A file containing Python code with functions, classes, and variables

Question 2

What is a package in Python? a) A single Python file b) A collection of related modules organized into directories c) A built-in Python library d) A type of function

Answer: b) A collection of related modules organized into directories

Question 3

Which keyword is used to import a module in Python? a) include b) require c) import d) load

Answer: c) import

Question 4

What is the correct syntax to import a specific function from a module? a) import function from module b) from module import function c) import module.function d) get function from module

Answer: b) from module import function

Question 5

What does the `as` keyword do when importing modules? a) It imports all functions from the module b) It creates a backup of the module c) It gives the module or function an alias/alternative name d) It checks if the module exists

Answer: c) It gives the module or function an alias/alternative name

Question 6

What does `from math import *` do? a) Imports only the math module b) Imports all functions and variables from the math module c) Creates a copy of the math module d) Deletes the math module

Answer: b) Imports all functions and variables from the math module

Question 7

Why is using `from module import *` not recommended for larger modules? a) It's too slow b) It uses too much memory c) It can cause potential namespace conflicts d) It doesn't work with large modules

Answer: c) It can cause potential namespace conflicts

Question 8

How do you create a custom module in Python? a) Use the `create_module()` function b) Write functions, classes, or variables in a Python file c) Import the `module_creator` library d) Use the `new_module` keyword

Answer: b) Write functions, classes, or variables in a Python file

Question 9

What file is needed in a directory to make it a Python package? a) `package.py` b) `init.py` c) `init.py` d) `setup.py`

Answer: c) `init.py`

Question 10

How do you import a module from a package? a) `import package.module` b) `from package import module` c) Both a and b are correct d) `import package->module`

Answer: c) Both a and b are correct

Answer Key

1. b
2. b
3. c
4. b

- 5. c
- 6. b
- 7. c
- 8. b
- 9. c
- 10. c

video 3

Question 1

What is Object-Oriented Programming (OOP)? a) A way to organize files in Python b) A programming paradigm based on the concept of objects containing data and code c) A method to handle errors in Python d) A technique for optimizing code performance

Answer: b) A programming paradigm based on the concept of objects containing data and code

Question 2

What are the four main principles of OOP mentioned in the video? a) Encapsulation, Abstraction, Inheritance, Polymorphism b) Classes, Objects, Methods, Functions c) Variables, Functions, Loops, Conditions d) Import, Export, Define, Call

Answer: a) Encapsulation, Abstraction, Inheritance, Polymorphism

Question 3

What is a class in Python? a) A built-in function b) A blueprint or template for creating objects c) A type of variable d) A module in Python

Answer: b) A blueprint or template for creating objects

Question 4

What is the purpose of the `__init__` method in a class? a) To delete objects b) To initialize/set up the object's initial state when created c) To print object information d) To copy objects

Answer: b) To initialize/set up the object's initial state when created

Question 5

What does `self` represent in a class method? a) The class name b) A reference to the current instance of the class c) The method name d) A built-in Python keyword

Answer: b) A reference to the current instance of the class

Question 6

What are attributes in OOP? a) Functions that define object behavior b) Variables that hold data specific to an object c) Keywords used to create classes d) Methods for accessing objects

Answer: b) Variables that hold data specific to an object

Question 7

What are methods in OOP? a) Variables that store object data b) Functions defined within a class that describe object behaviors c) Ways to import modules d) Techniques for error handling

Answer: b) Functions defined within a class that describe object behaviors

Question 8

How do you access attributes and methods of an object? a) Using square brackets `[]` b) Using parentheses `()` c) Using dot notation `(.)` d) Using curly braces `{}`

Answer: c) Using dot notation `(.)`

Question 9

What is the difference between instance variables and class variables? a) There is no difference
b) Instance variables are unique to each object, class variables are shared across all instances
c) Class variables are unique to each object, instance variables are shared
d) Both are the same thing with different names

Answer: b) Instance variables are unique to each object, class variables are shared across all instances

Question 10

In the Dog class example, what would `jackie.bark()` return if jackie's name is "Jackie"? a) "Jackie" b) "bark" c) "Jackie says woof" d) "Dog barks"

Answer: c) "Jackie says woof"

Answer Key

1. b
2. a
3. b
4. b
5. b
6. b
7. b
8. c
9. b
10. c

Chapter 11

Title :- Libraries in Python

Video1

Question 1

What is inheritance in Python? a) A way to import modules b) A mechanism that allows one class to inherit properties and methods from another class c) A method to handle errors d) A technique for organizing files

Answer: b) A mechanism that allows one class to inherit properties and methods from another class

Question 2

In inheritance, what is the class that provides properties and methods called? a) Child class b) Derived class c) Parent class or Base class d) Sub class

Answer: c) Parent class or Base class

Question 3

How do you create a child class that inherits from a parent class in Python? a) class Child(Parent): b) class Child inherits Parent: c) class Child extends Parent: d) class Child <- Parent:

Answer: a) class Child(Parent):

Question 4

What is method overriding? a) Creating new methods in a child class b) Deleting methods from a parent class c) Redefining a method in a child class that already exists in the parent class d) Calling multiple methods at once

Answer: c) Redefining a method in a child class that already exists in the parent class

Question 5

What is multi-level inheritance? a) A class inheriting from multiple classes simultaneously b) A chain-like hierarchy where a class inherits from another class, and a third class inherits from the second class c) Inheriting methods but not attributes d) Creating multiple instances of the same class

Answer: b) A chain-like hierarchy where a class inherits from another class, and a third class inherits from the second class

Question 6

What is the purpose of the `super()` method? a) To create a new class b) To delete a parent class c) To allow a derived class to call methods from its parent class d) To override all methods in a class

Answer: c) To allow a derived class to call methods from its parent class

Question 7

When is the `super()` method commonly used? a) To create new attributes b) In the constructor (`__init__`) method to initialize the parent class c) To delete objects d) To import modules

Answer: b) In the constructor (`__init__`) method to initialize the parent class

Question 8

What are the main benefits of inheritance? a) Faster code execution b) Code reusability, simplified maintenance, and improved code structure c) Better error handling d) Reduced memory usage

Answer: b) Code reusability, simplified maintenance, and improved code structure

Question 9

If a child class has a method with the same name as a method in the parent class, what happens?

a) An error occurs b) Both methods are called simultaneously c) The child class method overrides the parent class method d) The parent class method is always used

Answer: c) The child class method overrides the parent class method

Question 10

Why is `super()` preferred over directly referencing the parent class by name? a) It's faster b) It makes code more maintainable and easier to change c) It uses less memory d) It's required by Python syntax

Answer: b) It makes code more maintainable and easier to change

Answer Key

1. b
2. c
3. a
4. c
5. b
6. c
7. b
8. b
9. c
10. b
- 11.

video 2

Question 1

What is error handling in Python? a) A way to create errors in programs b) The process of responding to and managing errors or unexpected events in a program c) A method to delete error messages d) A technique for writing faster code

Answer: b) The process of responding to and managing errors or unexpected events in a program

Question 2

What is the main purpose of error handling? a) To make programs run faster b) To prevent programs from crashing and provide ways to respond gracefully c) To create more complex code d) To reduce memory usage

Answer: b) To prevent programs from crashing and provide ways to respond gracefully

Question 3

What is the basic syntax for error handling in Python? a) try-catch blocks b) try-except blocks c) catch-finally blocks d) error-handle blocks

Answer: b) try-except blocks

Question 4

What type of error occurs when trying to divide a number by zero? a) TypeError b) ValueError c) ZeroDivisionError d) IndexError

Answer: c) ZeroDivisionError

Question 5

What happens in the `finally` block? a) It only executes if an error occurs b) It only executes if no error occurs c) It always executes regardless of whether an error occurs or not d) It never executes

Answer: c) It always executes regardless of whether an error occurs or not

Question 6

What does the `raise` keyword do in Python? a) It handles exceptions automatically b) It allows developers to intentionally trigger exceptions under certain conditions c) It prevents exceptions from occurring d) It logs error messages

Answer: b) It allows developers to intentionally trigger exceptions under certain conditions

Question 7

What type of error occurs when you pass the wrong data type to a function? a) ZeroDivisionError b) ValueError c) TypeError d) IndexError

Answer: c) TypeError

Question 8

How do you create a custom exception in Python? a) By importing the exception module b) By subclassing the Exception class c) By using the `custom_exception()` function d) By defining a regular function

Answer: b) By subclassing the Exception class

Question 9

What is a best practice when raising exceptions? a) Use exceptions for regular control flow b) Raise as many exceptions as possible c) Use specific exception types and provide meaningful error messages d) Never use custom exceptions

Answer: c) Use specific exception types and provide meaningful error messages

Question 10

In the try-except structure, what happens when an exception is raised in the try block? a) The program continues executing the try block b) The program immediately stops executing the try block and jumps to the appropriate except block c) The program crashes d) The exception is ignored

Answer: b) The program immediately stops executing the try block and jumps to the appropriate except block

Answer Key

1. b
2. b
3. b
4. c
5. c
6. b
7. c
8. b
9. c
10. b

video 3

Question 1

What is Django? a) A database management system b) A high-level Python web framework for rapid development of secure and scalable web applications c) A Python library for mathematical computations d) A data analysis tool

Answer: b) A high-level Python web framework for rapid development of secure and scalable web applications

Question 2

What does ORM stand for in Django? a) Online Resource Management b) Object Relational Mapping c) Organized Resource Module d) Open Resource Manager

Answer: b) Object Relational Mapping

Question 3

What is the main advantage of using ORM in Django? a) It makes websites load faster b) You don't have to write raw SQL queries; Python functions are translated into SQL queries c) It reduces memory usage d) It makes the code shorter

Answer: b) You don't have to write raw SQL queries; Python functions are translated into SQL queries

Question 4

What does MTV stand for in Django's architecture pattern? a) Model Template View b) Module Template Variable c) Method Template Value d) Model Type View

Answer: a) Model Template View

Question 5

What does NumPy stand for? a) Number Python b) Numerical Python c) New Python d) Numeric Python

Answer: b) Numerical Python

Question 6

What is the core object in NumPy? a) list b) array c) nd-array (n-dimensional array) d) matrix

Answer: c) nd-array (n-dimensional array)

Question 7

Why is NumPy preferred over traditional Python lists for handling large datasets? a) It uses less memory b) It's easier to write c) It's significantly faster and more efficient d) It has better error handling

Answer: c) It's significantly faster and more efficient

Question 8

What is Pandas primarily used for? a) Web development b) Data analysis and manipulation c) Game development d) Network programming

Answer: b) Data analysis and manipulation

Question 9

What is the main data structure in Pandas? a) Series b) Array c) DataFrame d) Matrix

Answer: c) DataFrame

Question 10

What is an API in the context of web development? a) A type of database b) A programming language c) Application Programming Interface - a bridge between frontend and backend/database d) A web browser

Answer: c) Application Programming Interface - a bridge between frontend and backend/database

Answer Key

1. b
2. b
3. b
4. a
5. b
6. c
7. c
8. b
9. c
10. c